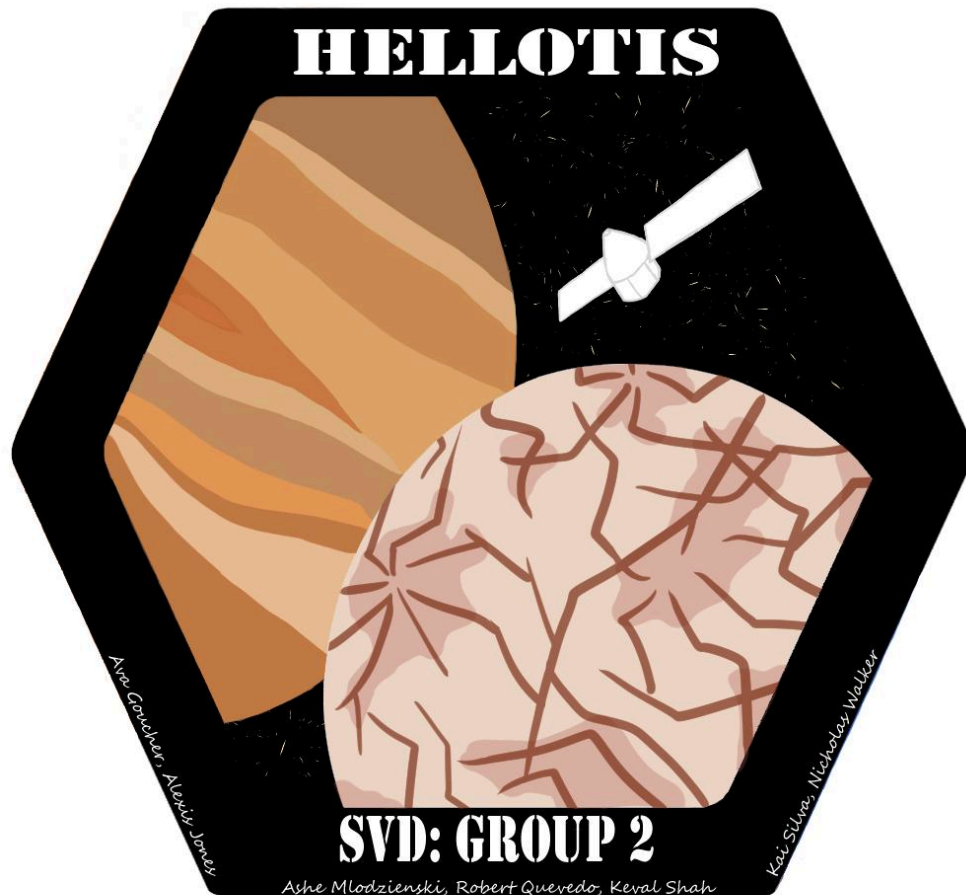


Group 2 Preliminary Design Review

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Executive Summary (Ashe)

For a mission to the icy Jovian moon of Europa, many preliminary decisions must be made with respect to the various subsystems within the Preliminary Design Review for the Hellotis Mission. For the structures, backbone truss with a hexagonal cylinder subsystem has been selected to be made out of Aluminum-7075. For the propulsion system, a dual-mode propulsion system has been chosen to power the VEEGA maneuver as well as both a sun-sensor and star tracker for ADCS. For power, a set of solar panels will be designed to fold in for launch on a Falcon 9 Heavy rocket. The design also will utilize Lithium-Ion batteries for storage, both active and passive thermoregulators, and various deployment mechanisms throughout the duration of the mission. Finally, to get information return, a centralized computer system will be used for data storage and a parabolic HGA will be used for both deep space transmission and an assortment of smaller MGA and LGA for lander communication.

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1. Introduction (Keval)

The question of whether life exists outside of Earth has sought to be answered long before space exploration was even possible. The search for life later became the search for water, as water is a key component to sustaining life. This search proved successful as a discovery in the 1960s revealed Europa's surface is mostly water ice. Following images from the Voyager missions, hints appeared that at some point in the past, Europa may have had a warm interior. This then led to the question if its surface composition allowed the surface to still contain an ocean of life underneath the icy surface. This prompted the motivation to send a spacecraft out to Europa and take a closer look.

The proposed Hellotis spacecraft intends to travel to Jupiter and analyze Europa during several fly-bys. Hellotis will also deploy the Minos probe onto Europa's surface to obtain samples and analyze the surface composition even further. Hellotis will contain several instruments including cameras, spectrographs, magnetometers, and radar systems to investigate surface properties and answer questions about the history of Europa. The Hellotis Design Team will design and analyze the production of a spacecraft capable of completing this mission efficiently.

As a reference to guide the general design of Hellotis, the Juno and Europa Clipper spacecraft will be studied and analyzed. These are the two most recent spacecraft that have traveled to Jupiter so the design choices made here will be more influential than older spacecraft, such as the Voyager missions. The design of Hellotis will aim to match and improve on the design of these spacecraft.

2. Customer needs (Ashe)

The purpose of the mission is to design a spacecraft that will orbit Europa to collect information on its surface composition using devices present on the orbiter and with the Minos probe that will land on Europa's surface. The spacecraft needs to survive to launch and carry the necessary instrumentation to return adequate information about Europa. The focus of the Hellotis Design Team is on the orbiter spacecraft. To begin the design process, the mission objectives, constraints, and assumptions must be made clear.

2.1. Mission Objectives

The following goals are set for the Hellotis mission:

- Determine the composition of Europa's surface
- Perform a fly-by to deploy the Minos probe
- Investigate the ocean underneath the icy surface
- Characterize Europa's atmospheric interactions with Jupiter

The achievement of these goals will quantify the success of the Hellotis mission. To achieve these goals, the orbiter first needs to be launched off of Earth's surface and then captured in an orbit around Jupiter. This will allow for the fly-bys of Europa and the analysis to be completed.

2.2. Mission Constraints

2.2.1. Technical Constraints

The main constraints are on size and mass. As the spacecraft will utilize the Falcon Heavy to launch, a high bound on the spacecraft weight will be 16000 lbs [22]. However, a more reasonable limit will be the mass of the Europa Clipper, which is 13000 pounds. The size of the spacecraft needs to be accommodated to be able to hold the Minos probe and all of the necessary instrumentation for deployment of the probe and analysis of Europa. Other constraints will be discussed in more detail as they pertain to each subsystem.

2.2.2. Time Constraints

The main time constraint for the mission is the time of launch. The trajectory uses gravity assists from Venus and Earth as well and a gravity capture at Jupiter. To ensure the accuracy of the trajectory, there is a specific launch window that must be met. If this launch window is not met, a secondary launch window must be defined. Similar to the Europa Clipper and Juno mission's the spacecraft is estimated to take approximately 6 years until the gravity capture at Jupiter.

2.2.3. Environmental Constraints

The major environmental factor affecting the mission design is the radiation from Jupiter. The planet's magnetic field is 20000 times as strong as Earth's and spins faster, completing an Earth day in only 10 hours [23]. This constraint requires a highly elliptical orbit around Jupiter to minimize the close range effects of radiation. Another environmental constraint is minimizing space debris left over after mission completion. NASA-STD-8719.14C provides the process for limiting orbital debris.

2.2.4. Political Constraints

As Hellotis will be launched as an American spacecraft, the mission must adhere to the agreements set in the Outer Space Treaty. Any violations of the standards will result in political consequences against the United States.

2.2.5. Cost Constraints

This mission does not have a “real” cost constraint, but to simulate a true design of a space vehicle, the goal will be to minimize costs of materials in designing Hellotis. As a benchmark, comparisons will be made to the Europa Clipper and the Juno spacecraft to determine how efficiently the team designed for the mission.

3. Design Details (Ashe)

3.1. Launch Vehicle

When designing a deep-space mission, it is imperative to have an ideal start to the mission, which is wholly defined by the launch vehicle and its capabilities. In this sphere of the design, choosing a launch vehicle is defined by the amount of mass it can take as well as how much volume that mass can take up. Additionally, only the heavy lift vehicles from various suppliers can be considered due to the amount of hardware and propellant required to reach the outer solar system.

The Falcon Heavy was selected for its highest mass to orbit capability of 16,800 kg to Trans-Mars Injection [19]. Because the mission is not stopping at Mars, the true capability is slightly higher, but this value provides an initial boundary value for the total mass.

3.2. Mass Budget

As given by the launch capacity of the Falcon Heavy, the mass budget for this project is 16,800 kg. This launch capacity must include the dry mass of the spacecraft as well as the additional propellant and lander masses.

3.2.1. Lander

Because the lander design has been delegated to other project teams, this project must make assumptions around the mass of the Minos Probe. As NASA has sent missions dealing with ice to Mars, this project will assume that the Minos Probe will be similar to the Phoenix Lander, which landed on the red planet in May of 2008 [20]. Any major differences between the Phoenix and Minos Landers will contribute to a

maximum of a 20% increase from the Phoenix Lander, yielding a total mass for the Minos Probe of 800 kg.

3.2.2. Propellant

Because the mission has a total lifetime of over 7 years from Earth departure to Jovian plunge, the amount of propellant is integral to the success of this mission. Due to this reliance on propulsion, the propellant has a total mass budget of approximately 50% of the total mass of the spacecraft, yielding a total mass for the propellant of 8000 kg.

3.2.3. Major Subsystems

After the lander and the propellant, the rest of the spacecraft has a total mass of 8000 kg. The Telecommunications subsystem's mass budget is approximately 35 kg for the HGA and array of other data collection devices [29]. The Command and Data subsystem's mass budget is approximately 300 kg when accounting for the electronic vault necessary for proper function [30]. Around 40% of the spacecraft's weight will be allocated to the structure subsystem. This gives the structures a mass budget of approximately 3200 kg. The Power subsystem's mass budget should account for approximately 600 kg, to be used for both the solar arrays and lithium batteries. The Thermal subsystem's mass budget is approximately 50 kg, to account for the various thermoregulators. All other subsystems have smaller budgets which will fill in the rest of the mass, it is expected that all subsystem will give a 5% margin on their exact numbers.

3.3. Volume Budget

The Falcon 9 Heavy Fairing is a cylinder with a diameter of 5.2 meters and a height of 18.7 meters [19]. This is the maximum volume that the spacecraft can fit within during its folded state. Because of this, the central body of the structure will fit within the circular cross section of diameter 4.2 meters, allowing for the folded solar panels to have a folded volume constraint of 0.5 meters radially. For the central body of the structure, an annular volume with a thickness of 0.1 meters is reserved for 14 meters vertically. Below the central body, 2 meters of height is reserved for the Minos Probe. Above the central body, the last 2 meters of height is reserved for the telecommunication system. Finally, within the central 4.0 meter diameter cavity within the exterior structure, the propellant tank will sit alongside the electronics vault, the battery storage, and the thrusters.

4. Structures (Keval)

4.1. Structures requirements

The structures subsystem involves selecting a design that can securely support all major components, both inside and outside the spacecraft, while meeting mission demands. An effective structural design not only achieves this but does so with minimal mass. Key design factors include the structure's mass, its strength under expected mission stresses, and its efficient use of volume. As the foundation of the spacecraft, these elements are crucial to mission success and must be carefully balanced to ensure optimal performance.

4.1.1. Launch Stresses

The structure will withstand all stresses from the launch. This is done by applying the maximum expected launch stresses to a finite element model of the spacecraft with simplified assumptions about exact mass distribution.

4.1.2. Fairing Connections

The structure must be sufficiently small such that it can be launched with the chosen launch vehicle while folded up. Additionally, the spacecraft must attach to the provided power and other connection ports specified by the launch vehicle's user guide.

4.1.3. Subsystem Connections

The structure must be built to securely and safely hold all required equipment related to the mission. This includes having connection points for all telecommunication, power, and propulsion modules.

4.1.4. Outer Space Environmental Effects

The structure must be built to withstand the space environment where ionizing radiation and micrometeorite impacts can jeopardize the mission.

4.1.5. Minos Probe Deployment Stresses

The structure must make sure of stability before, during, and after the deployment of the Minos Probe. Additionally, the shocks from this action must not affect the accuracy of the ADCS.

4.2. Structural Materials

An important factor when considering the structures of a spacecraft is the material that will be used to make it. Several key factors such as Tensile yield strength, density, cost etc.. were researched on different materials to create a material properties

table. These criteria help determine the optimal material for a mission. The most commonly used materials for spacecraft structures include metal alloys like aluminum, titanium, and other heat-resistant alloys, as well as composite materials like carbon fiber [9]. Table 1 provides several material options along with their associated physical properties.

Material	Tensile Yield Strength (MPa)	Density (g/cm ³)	Thermal Conductivity (W/m·K)	Cost (USD/kg)	Max Elongation (%)	Ease of Manufacture
7075 Aluminum	503	2.81	130	2.5 – 3.0	8	Medium
6061 Aluminum	276	2.7	167	2.0 – 2.5	12	High
Titanium	880	4.43	22	30 – 50	14	Difficult
Kevlar	3620	1.44	0.04	50 – 60	2	Medium
Carbon Fiber	3500	1.75	5	80 – 120	1.5	Difficult
Steel (A36)	250	7.85	50	0.5 – 1.0	25	High
Beryllium	345	1.85	200	450 – 800	2.5	Very Difficult
Inconel (718)	1034	8.19	11	30 – 80	30	Difficult

Table 1: Material Properties for Possible Material Selections [10, 11, 12, 13, 14, 15, 16, 17]

A decision matrix was then made to choose the material best suited for the mission. This matrix represents. The matrix focused primarily on strength and density, as these are crucial for the mission's success and lander design. Temperature tolerance was the next significant factor due to the extreme conditions the structure will endure. Although cost and manufacturability were also taken into account, they were given less priority since they can be addressed if the material meets the more critical performance requirements.

	Weight	Aluminum-7075	Aluminum-6061	Titanium	Carbon Fiber	Kevlar	Inconel-718
Tensile Yield Strength	3	2	1	4	5	5	3
Density	3	4	4	3	5	5	1
Thermal Conductivity	2	5	5	3	1	1	2
Cost	0.5	5	5	3	2	2	2
Max Elongation	1	3	3	4	1	1	5
Ease of Manufacturing	0.5	5	5	1	1	2	1
Total	50	36	33	33	34.5	35	22.5

Table 2: Decision Matrix for Material Selection

From the decision matrix Aluminum 7075 was the material that was chosen as the material to be used for the orbiter. Aluminum 7075 is an aerospace-grade material that is commonly used in spacecraft. The Europa Clipper spacecraft also primarily uses Aluminum 7075 for its structural components. This is due to its high strength to weight ratio, excellent fatigue resistance, and suitability for the demanding conditions of space exploration. Additionally, 7075 is known for being stronger than Aluminum 6061, though it is slightly more expensive and more difficult to work with. The trade-off in favor of strength makes it ideal for spacecraft structures, which must withstand high loads during launch and survive in extreme environments.

4.3. Structure shape

In addition to material selection, the shape of the structure was also considered. Various shapes have been used in previous satellites, including spheres, cylinders, octagons, and others, depending on the mission requirements. For the orbiter, four primary shapes were evaluated: cylinder, octagon, backbone truss, and hexagon. A decision matrix was created using specific criteria to assess and compare these shapes for suitability.

Criteria	Cylinder	Octagon	Backbone Truss	Hexagon
Mass	Efficient for volume	Low	Lightweight	Good
Strength	Moderate	Strong	Very Strong	Very Strong
Use of Volume	Good	Efficient	Excellent	Efficient
Accessory Mounts	Complex	Easy	Easy	Excellent
Cost	Low	Moderate	Cost-effective	Higher

Table 3: Criteria for the Structure Shape

A decision matrix was then made to choose the best shape for the EO. The matrix weighted mass and strength heavily, this was because of these crucial points for the mission. The shape that was chosen will be a Truss, as seen it is the best candidate in the matrix.

	Weight	Cylinder	Octagon	Backbone Truss	Hexagon
Mass	3	5	1	2	3
Strength	3	3	4	5	5
Use of Volume	2	3	4	5	4
Accessory Mounts	1	1	2	5	5
Cost	1	4	4	5	2
Total	50	35	29	41	39

Table 4: Decision Matrix for Internal Spacecraft Shape

A backbone truss is great because it offers a lightweight, strong, and stiff framework that efficiently supports loads. Backbone truss configurations, like triangular or square trusses, make it easy to mount components securely, allowing for straightforward routing of spacecraft systems such as wiring, propulsion lines, and thermal controls [18]. This structure is also easier to manufacture and maintain, as components are more accessible. The structure will be a hexagonal cylinder structure. This will allow for a mounting of the propellant, vault, and the minos lander.

4.4. External Skin

A thin-walled honeycomb structure will be chosen by the team for the satellite due to its ability to provide an optimal balance between low mass and high stiffness, which is essential for reducing weight while maintaining structural integrity. The honeycomb core, positioned between two thin sheets, increases resistance to buckling under compressive loads, addressing a common issue in lightweight structures. Additionally, this construction method improves the satellite's durability and performance by allowing local loads, such as those at attachment points, to be efficiently distributed, preventing material tearing or crushing. The lightweight and strong properties of the honeycomb structure make it well-suited for spacecraft, particularly in high-temperature environments. As of now the satellite does not need a skin due the use of a backbone structure and the electronics being in the vault., but in the future if a skin is needed for additional protection the honeycomb design will be used.

4.5. Vault

Jupiter has an intense radiation belt similar to Earth's Van Allen belts, which the orbiter will be flying through. This raises concern that the already radiation hardened electronics will not be able to survive the zone. In order to mitigate the risk of losing important electronics, a "vault" will be designed. This vault will utilize a material that

will protect the internals from Jupiter's radiation belt. The vault will be in a rectangular prism shape, very similar to the one used in NASA's Europa Clipper. The prism will house all crucial electronics and be mounted to the truss structure.

4.6. Preliminary Design Idea

Using the above conclusions found in each decision matrix, the team constructed a preliminary structural design of the Europa Orbiter. Figure 1 displays the trimetric CAD model of the structure. The model will be updated and refined as future work is carried out. Within the model, the black box is the electronics vault, the gold box is the lander, the red box is the propellant tank, the tan box is the telecommunications module, and the blue is the solar panels. Additional components will take up less volume and will be modeled in future iterations.

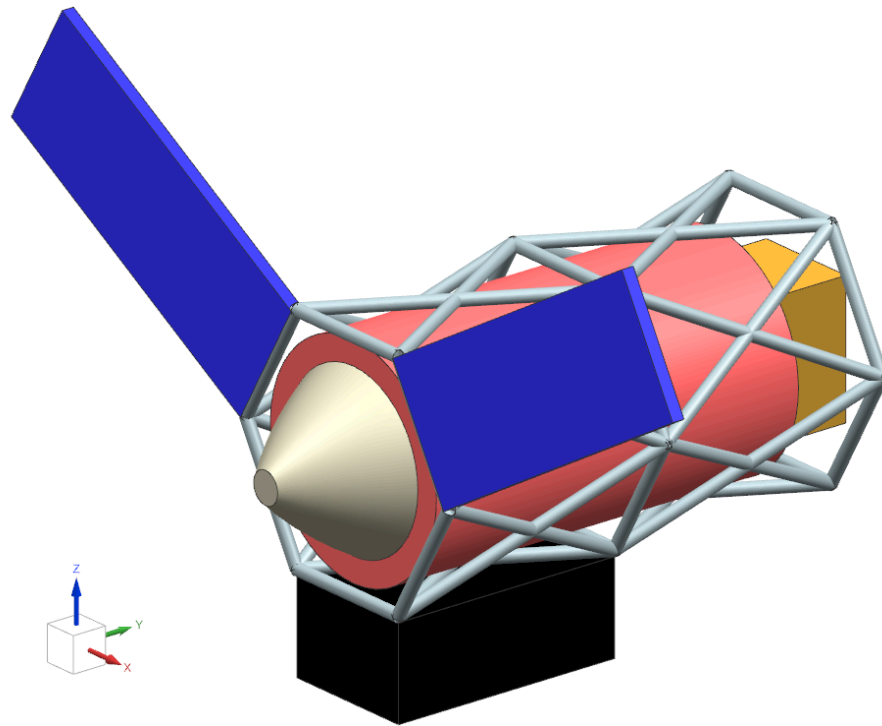


Figure 1: Preliminary Design CAD Model.

4.7. Risk Mitigation

Risk mitigation is a critical aspect of the structural subsystem, as any failure could lead to mission failure. Ensuring the structure can endure the harsh conditions of space, such as extreme radiation and mechanical stresses, is essential. Identifying potential risks early on allows for effective preventative measures to be implemented,

minimizing the chances of structural failure. By focusing on these risks and their mitigations, the team can enhance mission reliability and longevity. Table 5 below provides an overview of the primary risks related to the structural subsystem and the strategies in place to manage them.

Risk	Cause	Mitigation	Severity
Outgas	Escaping gasses causes potential issues with sensitive equipment	Conduct comprehensive outgassing tests during ground testing	3C (Major, Remote)
Thermal cycling	Repeated thermal expansion and contraction damaging materials or components	Use materials or coatings designed to withstand space's thermal extremes	2B (Hazardous, Improbable)
Resonance with launch vehicle	Structural resonance with the launch vehicle causes vibration damage	Perform vibration analysis and damping measures to prevent structural damage during launch	4B (Hazardous, Occasional)
Radiation	Jupiter's radiation field could cause failures in the electronics	Create a radiation protection compartment for the electronics to be stored.	5A (Catastrophic, Frequent)

Table 5: Technical Risks for the Spacecraft Structures

4.8. Non Technical considerations

In addition to the risk factors, the team has also taken into account a range of non-technical factors for this subsystem. These non-technical considerations are outlined below in Table 6.

Topic	Consideration
Public Health & Safety	This design ensures that the structure will not fracture around Earth or at impact, mitigating concerns posed to the general public due to reentry.
Global	N/A - The structural subsystem does not have any significant

	global considerations as it is unlikely to affect change on the global level.
Cultural	N/A - The structural subsystem does not have any significant cultural considerations as it is unlikely to affect change on the cultural level.
Social	N/A - The structural subsystem does not have any significant social considerations as it is unlikely to affect change on the societal level.
Environmental	The materials used will be sourced from a manufacturer that obtains them in an environmentally friendly manner.
Economic	The materials used may be expensive and difficult to source, benefiting the companies selected to provide the materials economically.
Professional Ethics	The designer has completed relevant courses and has industry experience. All services and statements will be truthful to the extent of the team's knowledge.

Table 6: Non-Technical Risks for the Spacecraft Structures

4.9. Future Work

In future work, the team will focus on selecting a material that can shield the electronic components from Jupiter's intense radiation. This material will form a protective skin around the most critical systems. Additionally, a finite element analysis (FEA) will be conducted on the structure to ensure it can endure the various loads and stresses expected during the mission. The team will also perform vibration analysis to assess the structure's ability to withstand shocks and vibrations during launch and orbital maneuvers.

Due to time limits and funding, some of the work required to continue this design cannot be performed. For example building a full scale model of the structure to perform physical tests. The team will be only conducting FEA analysis on a digital model. This will allow for testing of the spacecraft as whole, and allow the team to make sure the structure is safe for launch.

5. Orbital Mechanics (Ashe)

5.1. Trajectory to Jupiter

One goal of the Orbital Mechanics subsystem is the definition of the trajectory from Earth to Europa. Many trajectories have been looked at including Gravity Assisted, Low-Thrust, and high energy Hohmann transfers with the following five trajectories being selected as possibilities for the mission.

5.1.1. Europa Clipper MEGA Trajectory

The Mars-Earth Gravity Assist (MEGA) trajectory is being utilized by the Europa Clipper Mission and utilizes two gravity assist flybys before insertion into Jupiter orbit. Utilizing the analysis done by Petropoulos et al., possible launch hyperbolic velocity of 5.50 km/s yields a final capture hyperbolic velocity of 5.60 km/s [1]. As shown by figure 2, the flybys occur with Mars in February of 2025 and with Earth in December of 2026 before final insertion at Jupiter in 2030.

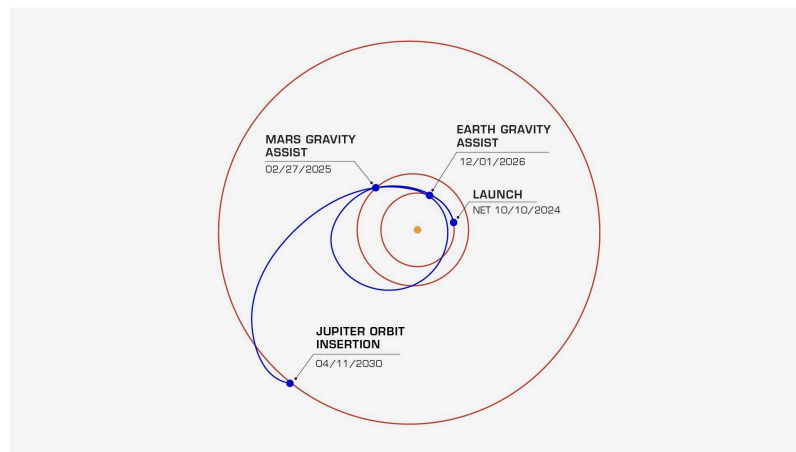


Figure 2: Europa Clipper Flight Trajectory with Flyby Dates [2]

5.1.2. Galileo VEEGA Trajectory

The Venus-Earth-Earth Gravity Assist (VEEGA) trajectory was used by the Galileo spacecraft, a mission which launched in 1989 and utilized three gravity assist flybys before insertion into Jupiter Orbit [3]. Because this ballistic trajectory was done almost 40 years ago, an updated trajectory design is required.

Analyzed by Petropolus et al., a possible trajectory was designed for a launch in November of 2029 with a launch hyperbolic excess velocity of 3.25 km/s yielding a final capture hyperbolic velocity of 6.40 km/s [1]. In this trajectory, the Venus flyby occurs in March of 2030, the Earth-1 flyby occurs in February of 2031, and the Earth-2 flyby occurs in May of 2033 before final insertion in June of 2035.

5.1.3. VEME Gravity Assist Trajectory

The Venus-Earth-Mars-Earth Gravity Assist (VEMEGA) trajectory is a hypothetical ballistic trajectory developed by Petropolis et al. and utilizes a relatively low initial launch hyperbolic excess velocity to get to Jupiter in 6.4 years [1]. This trajectory uses four gravity assists, two of which around earth: allowing for any number of possible flyby missions.

The possible trajectory was designed for a launch in May of 2023 with a launch hyperbolic excess velocity of 3.25 km/s yielding a final capture hyperbolic velocity of 5.58 km/s [1]. In this trajectory, the Venus flyby occurs in April of 2023, the Earth-1 flyby occurs in July of 2024, the Mars flyby occurs in December of 2024, and the Earth-2 flyby occurs in September of 2026 before final insertion in August of 2029.

5.1.4. Low-Thrust MGA Trajectory

The Low-Thrust Mars Gravity Assist (LTMGA) trajectory utilizes a specific kind of propulsion system, which applies smaller amounts of thrust over a larger amount of time. These systems are more efficient, but are much more complex than standard ballistic trajectories. This LTMGA trajectory was analyzed and developed by Saghmanesh et al. and uses an initial spacecraft mass of 19819 kg after the launch excess hyperbolic velocity of 0.8 km/s [4].

The trajectory shown below in figure 3 uses a specific impulse of 6000 s and has a maximum thrust value of 2.26 N. The trajectory has a launch data of February 2022 and has a single gravity assist from Mars in March of 2024 before insertion in February of 2027 with an excess hyperbolic velocity of 0.0 km/s.

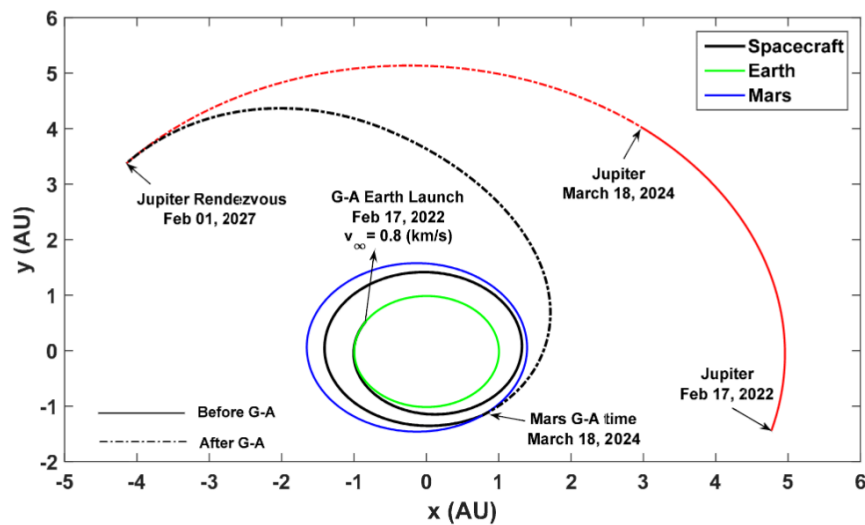


Figure 3: Low Thrust Mars Gravity Assist Trajectory with Flyby Dates [3]

As provided in the analysis, the final mass of the spacecraft is 15068 kg, which is heavy, but would allow for a longer mission about Jupiter and its moons.

5.1.5. Earth-Jupiter Hohmann Transfer

A Hohmann transfer is done as a very simple transfer orbit, which utilizes an ellipse between the Earth and Jupiter orbits. Because Jupiter is at an orbital radius of 5.2 AU and Earth is at 1.0 AU, the initial excess hyperbolic velocity is given as 8.80 km/s and the final excess hyperbolic velocity is given as 3.86 km/s for insertion for a total flight time of 2.72 years.

5.1.6. Decision Matrix

The following decision matrix takes into account the required ΔV to both initiate and complete the trajectory as well as the number of interplanetary flybys that could go wrong.

Within table 7, the low thrust maneuver was disqualified due to the required complexity of the propulsion system and the reworking of the entire spacecraft design around that system. Given the data shown, the project is moving forward with the Venus-Earth-Earth Gravity Assist trajectory outlined in section 6.1.3.

	Weight	MEGA	VEEGA	VEMEGA	Hohmann
Complexity of Trajectory	2	4	3	1	4
Initial Delta V	3	2	4	4	0
Final Delta V	3	3	2	3	2
Time of Flight	1	3	3	2	4
Total	50	27	28	26	19

Table 7: Decision Matrix for Interplanetary Trajectory. Higher totals show a more advantageous design.

5.2. Capture at Jupiter

The second goal of the Orbital Mechanics subsystem is the definition of captured orbits around the target. When capturing the spacecraft around Jupiter, there are a few simplifying assumptions needed to make any sort of preliminary capture analysis possible. Firstly, we can assume that the Jupiter-Spacecraft system is a two-body problem, which has an analytical solution. This can be done because the moons of Jupiter are very small and only affect the spacecraft during very close flybys.

Secondly, we can assume that we can control the point of intersection between the spacecraft and Jupiter's sphere of influence. This can be done because the flight time is so long that a fast majority of incoming conditions can be met.

When slowing down to enter into one of these orbits, there are a few ways to reduce the spacecraft's velocity. The easiest way to do this is a major body gravity capture. In this situation, the spacecraft uses the inertia of the main planetary body to eat up some of its speed in the heliocentric frame before using a ballistic counterburn. After this counterburn puts the spacecraft into a semi-stable orbit, small adjustments are made to establish the main orbit. Another way to get captured by the planet is a minor body gravity capture. In this situation, the spacecraft uses a series of smaller gravity capture movements to reduce the orbital energy of the spacecraft in the planetary frame before ending up in the same situation as the major body gravity capture. Finally, there is the process of aerobraking, where a calculated collision with the atmosphere of the major body allows for the use of aerodynamic drag to reduce the orbital energy of the spacecraft [5]. This requires highly accurate predictions of the planet's atmosphere as well as an internal control system to either increase the drag or increase the spacecraft's speed. Because of these requirements, it is not recommended while other options remain available.

5.2.1. Minimally Eccentric Orbits

A minimally eccentric orbit ($e < 0.1$) is an orbital family that has a relatively consistent orbital distance around the host body. For these orbits, the spacecraft could stay fairly close to Europa for its entire mission, only using small adjustments to keep close to the moon and be able to continuously receive data from the Minos probe.

There are two main problems with this orbital family. Firstly, a minimally eccentric orbit would cost a very large amount of propellant and thrust even with the velocity reduction maneuvers the spacecraft would take. This would reduce the amount of corrections that the spacecraft would be able to make, reducing mission lifetime. Secondly, Europa sits within Jupiter's magnetosphere, intense belts of radiation similar to Earth's Van Allen belts [6]. This radiation affects the long-distance communication ability for the orbiter and would damage the sensitive internal electronics if long term exposure happened.

5.2.2. Highly Eccentric Orbits

A highly eccentric orbit ($e > 0.5$) is an orbital family that has a varied orbital distance around the host body. For these orbits, the spacecraft would spend a majority of its mission time far from Europa, only performing flybys every 4-6 orbital periods.

This type of orbit has the major upside of a smaller capture impulse. This allows for longer missions and more chances for corrective burns as well as a smaller number of risky minor gravity capture maneuvers. Additionally, the highly eccentric orbits have a well defined set of mission phases. The first phase is during elliptic periapse, where scientific objectives can be met and data is collected as the spacecraft is closer to

Jupiter. The second phase is during elliptic apoapse, where the spacecraft has a more direct line of sight on earth.

This orbital family is not without issue, due to only having periodic flybys of the Minos probe. The probe would not be able to have a constant data stream to the orbiter and would need to provide that information in dense packets, which may be affected by the atmosphere of the moon

5.2.3. Constantly Adjusted Orbits

A constantly adjusted orbit uses many smaller but carefully timed velocity impulses to fine-tune the exact orbital parameters for each flyby. This has the upside of being able to visit many targets in each revolution around Jupiter and the ability to get into “perfect” position for each transmission between the probe, the orbiter and Earth.

Just like every other captured trajectory, there are some issues with an adjusted orbit. Firstly, the main issue of complexity. Having many small adjustments can make every impulse a possible point of failure as well as a use of fuel, which would reduce the length of the mission. Additionally, small adjustments are not easy to produce with standard propulsion systems, so many of the adjustments would be corrective adjustments of the anticipated adjustments.

5.2.4. Decision Matrix

Because the conservation of fuel is a major requirement for the gravity capture, it can be assumed that a minor body gravity capture will be used to arrest some of the spacecraft’s momentum. However, this does not mean that every type of semi-stable orbit is created equal. The following decision matrix highlights the major differences of the required complexity of the propulsion control system, the capture’s required ΔV , and the probability/severity of various risks stated in each category’s section above.

Within table 8, all three of the possibilities are shown and compared on a 1-5 scale with variable weights. Given the data, the project is moving forward with the highly eccentric capture orbit outlined in section 6.2.2

	Weight	Minimal Eccentric	Highly Eccentric	Adjusted
Complexity of Control	2	3	3	5
Required Delta V	3	2	5	3
Probability of Failure	3	3	5	3
Severity of Risk	2	4	4	2
Total	50	27	42	32

Table 8: Decision Matrix for Jovian Captured Trajectory. Higher totals show a more advantageous design.

5.3. Possible Jovian Moon Flybys

As a part of defining orbits about Jupiter, it is imperative to look at the various moons and define a set of possible flybys of those moons. Many of the moons are incredibly small, so it is important to get close up images during missions to the gas giant.

5.3.1. Galilean Moons

Jupiter's Galilean moons were discovered in 1610 and are the most readily visible objects around the planet. Io, Europa, and Ganymede are in a 1:2:4 orbital resonance, and are all tidally locked to the planet at various orbital distances. Because of this, it becomes easy to design an orbit that interacts with all three of the bodies. The fourth moon, Callisto, is further out and breaks the pattern, but it is still possible to add it to potential flybys.

Using the two body orbital equations for circular orbits, we can define the orbital parameters of each of the Galilean Moons. Io has an orbital radius of 421,800 km with a period of 1.769 Earth days. Europa has an orbital radius of 671,100 with a period of 3.551 Earth days. Ganymede has an orbital radius of 1,070,400 with a period of 7.155 Earth days. Callisto has an orbital radius of 1,882,700 with a period of 16.689 Earth days. These values will be taken into account for FDR analysis of the mission capture trajectory.

5.3.2. The Other Moons

Jupiter has a total of 95 moons, each with their own merits for a flyby. For simplicity, two additional moons are looked into from both a scientific perspective and from a feasibility of flyby standpoint.

The outer moon of Himalia was discovered in 1904 and has an orbital period of 251 Earth days. The main draw of Himalia is to look at one of the larger satellites in prograde orbit around Jupiter. Himalia has a radius of about 85 km, making it too small to majorly affect the orbit of the satellite, even in a close flyby. In addition to not costing much in the way of propulsion, the moon is theorized to have been an initial part of a large captured asteroid [7]. Analysis of this moon would allow for that theory to gain more empirical evidence.

The outer moon of Pasiphae was discovered in 1908 and has an orbital period of 708 Earth days. The main draw of Pasiphae is to look at a much smaller satellite in a retrograde orbit around Jupiter. Pasiphae has a radius of about 30 km and has a very large orbital period while still being one of the larger outer moons of Jupiter. Analysis

of this moon could give insight into the properties of the moon and why it has such a large deviation from the other larger outer moons.

5.4. Risks of Flight

Orbital Mechanics as a discipline is not without very strong risks. A 1 minute miss in the launch window could be the difference between successful flyby and catastrophic failure. What could go wrong!

5.4.1. Gravity Assist Risks

Gravity Assists are an inherently high risk and unforgiving maneuver. Due to the high velocities relative to the planet during an assist, even a small error in initial condition can make a trajectory fail as it can be nearly impossible to correct during the very short flyby window.

There are very few ways to correct for this error, but they all involve the system's guidance system, a working star tracker, and ground-based communication to apply small deep-space corrections before the assist takes place. Mitigation for this risk is a concrete plan of action and a reduction in the amount of total gravity assists during the flight trajectory.

5.4.2. Capture and Counterburn Risks

In addition to gravity assists, gravity capture maneuvers are equally as dangerous due to the risk of crashing into the capturing body. Additionally, a two-body model for a minor body gravity capture may not be sufficient due to the sheer mass of Jupiter.

The correction for this error is the same as for the gravity assists, but because the spacecraft is at the target, there is a very large time delay for any ground-based corrections. In this case, the capture and subsequent counterburn must be carefully timed and modeled before any issues can arise. Mitigation for this risk is preliminary capture models and on-board control for achieving necessary corrections.

5.4.3. Jovian Moon Flyby Risks

Finally, there is always a risk that there is an issue with the ground-based observations of the Jovian moons. They are comparatively very small around their host and the gravity well of Jupiter is not as well studied as Earth's or Mars'. This allows for the possibility that the flybys of each of the moons could provide risk to the spacecraft following its optimal trajectory. Mitigation for this risk is to allow a safety factor of used propellant for each planned trajectory in case of an unexpected increase or decrease in orbital energy after each flyby.

5.5. Additional Unanalyzed Concerns and Future Work

Outside of the risks to the mission shown above, there are a fair number of external variables that can jeopardize the mission due to the assumptions made about Jupiter as a host body for the spacecraft.

5.5.1. Spherical Harmonics

Spherical Harmonics are a byproduct of a non-spherically-symmetric distribution of the mass of a host body of an orbital system. These spherical harmonics act as a perturbation to the two-body system and rotate a stable orbit about the center of the host body. This can be either a help or a hindrance if this precession provides a useful change in the orbital elements.

The main spherical harmonic utilizes the J_2 value of the host. As shown by the data provided by the JUNO spacecraft, this value is 14696.514 with an uncertainty of 0.272 [8]. This parameter and its effects will be taken into account for FDR analysis of the mission capture trajectory.

5.5.2. Perturbation from Jovian Moons

The 4 largest Jovian moons; Io, Europa, Ganymede, and Callisto are massive enough to have a sphere of influence that is non-trivial. Because of this, their effects on a two-body trajectory can be quantified as a perturbing acceleration under the specific conditions of a flyby. Because this can be modeled ahead of time, this information will be taken into account for the determined flybys.

5.5.3. Perturbation from the Sun

Although the spacecraft will be spending all of its capture within the sphere of influence of Jupiter, the host planet spends its orbit within the sphere of influence of the sun. Although the effects of the sun will be minimal, the solar radiation pressure and the changes in distance can be analyzed as a perturbing acceleration during the entire mission. This will be looked into before FDR to show that it either will not affect the success of the mission or that the effects can be mitigated.

5.5.4. Jovian Magnetosphere

In addition to effects on the trajectory, the surroundings of Jupiter must be sufficiently analyzed so as to not jeopardize the mission. One of the largest issues with staying near Jupiter is the magnetosphere: intense belts of radiation similar to Earth's Van Allen belts [5]. These belts of radiation are very destructive to electronic equipment, which is vital to the success of the mission to both receive data from the Minos probe and to relay that information back to Earth through the Deep Space Network. Although radiation shielding is a component of physical design, a trajectory that

spends more time outside these radiation belts will allow for lighter shielding and more mass for scientific instruments, fuel for propulsion, and additional batteries.

5.6. Non-Technical Considerations

5.6.1. Public Health & Safety Considerations

Because Earth flybys are a component of the chosen trajectory, the risks of a failed gravity assist must be noted. Although mitigation has already been discussed, positions of earth-based satellites must be taken into account during the maneuver. If a failed gravity assist occurs, quick action must be taken to catch or destroy the orbiter before harm to the public occurs.

5.6.2. Global Considerations

No global concerns are considered due to the scale of the analysis discussed for this subsystem.

5.6.3. Cultural Considerations

No cultural concerns are considered due to the scale of the analysis discussed for this subsystem.

5.6.4. Social Considerations

No social concerns are considered due to the scale of the analysis discussed for this subsystem.

5.6.5. Environmental Considerations

Heavy considerations for the environmental safety of Europa must be taken into account. Because this mission may be life-discovering, extra care must be taken to not contaminate the moon or the samples taken. For the case of orbital mechanics, a plan for end of mission life must be developed to not contaminate any planet, moon, or asteroid. Similar to the Galileo mission, the Hellotis mission will end with a final plunge into Jupiter's high-pressure core [3].

5.6.6. Economic Considerations

No economic concerns are considered due to the scale of the analysis discussed in this subsystem.

5.6.7. Professional Ethics Considerations

The designer has completed relevant courses and has industry experience. All services and statements are truthful to the extent of the team's knowledge.

6. Propulsion (Robert)

6.1. System Definition

The purpose of the propulsion system is to provide the necessary thrust for the spacecraft to reach the Jupiter-Europa system and to aid in attitude control during the spacecraft lifetime. The two main functions the propulsion system needs to accomplish for Hellotis is to establish an orbit around Europa and to perform smaller maneuvers for attitude determination and control.

The burn will be accomplished with a larger thruster, while smaller boosters will be used for attitude control maneuvers.

6.2. Initial Selection

The main options for propulsion systems are cold-gas, solid propellant, liquid mono-propellant, liquid bi-propellant, liquid dual mode, and electric propulsion [18]. Electric propulsion has the highest specific impulse, going up to 3000 seconds, but provides very low thrust. Electric propulsion also requires heavy infrastructure and is very difficult to implement on a spacecraft. Due to these reasons, electric propulsion was not considered.

Table 9 shows the remaining propellants considered for the propulsion system and how they were assessed against the mission requirements. The factors considered in evaluation were weight, complexity, specific impulse, flexibility, throttling, and storage. Like all spacecraft, saving weight and minimizing complexity is always highly valued to have more room for structure or payload. These two factors weighed the highest. The next most important factor is specific impulse, as it is a measure of how effectively an engine uses its propellant. Storage is weighted high due to extra structure that may be needed for high-maintenance propellants. Flexibility concerns how the propulsion system handles different tasks, such as low thrust for attitude control and/or high thrust for orbit capture. Throttling accounts for how easily the amount of thrust can be changed as the propulsion system operates. This is especially important for low-thrust maneuvers. All of these factors are weighted and multiplied by the performance of each propulsion system (on a 1-5 scale, relative to the viable options) to select the propulsion system for Hellotis.

	Weight	Cold-Gas	Solid	Mono-Propellant	Bi-Propellant	Duel Mode
Weight	3	5	2	3	2	2
Complexity	1.5	5	4	4	3	3
Specific Impulse	2	1	4	2	5	4
Flexability	2	1	1	3	3	5
Storage	1.5	4	4	4	3	3
Total	50	32.5	28	31	31	33

Table 9: Decision Matrix for Propellant Type [21]

Based on the results, a dual mode propulsion system is selected. This will allow for both low-thrust maneuvers in attitude control and high-thrust maneuvers in allowing for gravity capture to establish the orbit combined within a single system. Despite the additional weight and complexity compared to the other systems, the dual-mode system has the capability of performing both of the propulsion system requirements in a single device. The most common chemicals used in dual mode propulsion are nitrogen tetroxide and hydrazine, as opposed to monomethylhydrazine and hydrogen peroxide. This combination was also used on the Juno spacecraft, proving its good selection as it has successfully reached an orbit around Jupiter. Hydrazine can be used by itself as a mono-propellant in low-thrust maneuvers. When combined with nitrogen tetroxide, it creates a bi-propellant for high-thrust maneuvers.

6.3. Preliminary Analysis

As a mono-propellant system, hydrazine is known to have a specific impulse of 220 seconds. As a bi-propellant system, hydrazine and nitrogen tetroxide is known to have a specific impulse of 339 seconds. A preliminary survey of the amount of propellant required can be done using the equation below[18].

$$m_p = m_0 [1 - e^{-(\Delta V / I_{sp} g)}]$$

where m_0 is the initial mass of the spacecraft. Knowing the specific impulse and acceleration due to gravity, we can conduct the survey using the assumption that Hellotis will be approximately 6065 kilograms, which is the mass of the Europa Clipper. Figure 4 displays the results of the bi-propellant system:

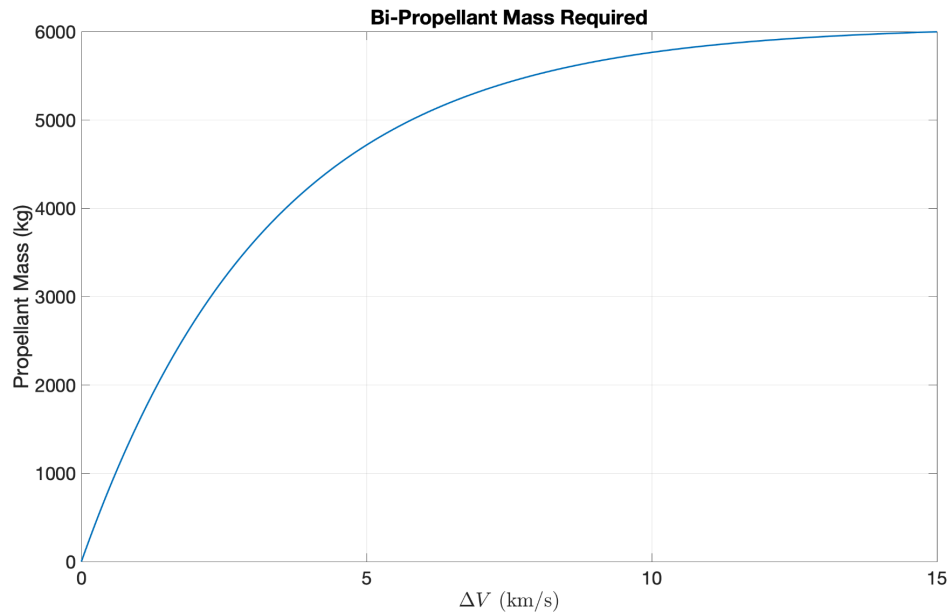


Figure 4: Initial Required Bi-propellant Mass Estimations

Once the appropriate delta-V is calculated, the mass of the bi-propellant will be known to size the structure of the propulsion system.

6.4. Risk Assessment

The dual mode is one of the more complicated propulsion systems as shown in table something, as it is combining the mono-propellant and bi-propellant systems into one. Table 10 outlines the possible risks associated with the propulsion systems and the actions to mitigate the risks. Severity of the risks is also weighted on a scale from 1-5.

Risk	Mitigation	Severity
Propellant instability due to perturbations during launch.	An adequate storage system will be created for both propellants in Hellotis.	4
Propulsion does not work due to the propellants' low freezing points.	An adequate thermal environment will be created for both propellants, ensuring they are within operating temperature range.	3
Propellant produces thrust when not called for.	The control system will be high-speed to ensure thrust maneuvers are performed without delay. A specific sequence will be required to initiate	2

	thrust so there are no accidental maneuvers.	
Run out of propellant after mission completion.	A factor of safety will be applied in the analysis to determine how much propellant will be brought.	1

Table 10: Technical Risks for the Spacecraft Propulsion

6.5. Future Work

A deeper analysis will be conducted alongside the orbital mechanics analysis to determine the delta-v required for the maneuvers to insert Hellotis into its desired orbit. The entirety of the propulsion system will be designed and optimized, including propellant tanks, engines, and flow control systems.

6.6. Non-Technical Considerations

Topic	Consideration
Public Health & Safety	Both hydrazine and nitrogen tetroxide are toxic to humans but extra care will be taken in handling these chemicals to prevent harm.
Global	N/A - The propulsion subsystem does not have any significant global considerations as it is unlikely to affect change on the global level.
Cultural	N/A - The propulsion subsystem does not have any significant cultural considerations as it is unlikely to affect change on the cultural level.
Social	N/A - The propulsion subsystem does not have any significant social considerations as it is unlikely to affect change on the societal level.
Environmental	The chemicals will be properly handled and disposed of properly in an environmentally friendly manner.
Economic	The materials used may be expensive and difficult to source, benefiting the companies selected to provide the materials economically.
Professional Ethics	The designer has completed relevant courses and has industry experience. All services and statements will be truthful to the

	extent of the team's knowledge.
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Table 11: Non-Technical Risks for the Spacecraft Propulsion

7. Attitude Determination and Control Systems (Ava)

7.1. System Definition

Attitude Determination and Control Systems (ADCS) is responsible for analyzing and managing the spacecraft's orientation including the spacecraft's solar panels, antennas, arms, etc. The angular motion and orientation of the spacecraft are monitored to guarantee the target attitude is maintained with respect to celestial bodies, stars, and planets. The success of the system's communication, power, and navigation rely on proper attitude control. [33]

The ADCS uses a variety of sensors and actuators; sensors track and collect data on the spacecraft's attitude with respect to the Sun, stars, and planets, while actuators take the data and analysis from the sensors to adjust the spacecraft's orientation as necessary for the desired attitude. Types of sensors include star trackers, inertial measurement units (IMU), sun sensors, and horizon sensors. Actuators include reaction wheels, reaction thrusters, gyros, and magnetorquers. [32]

7.2. Attitude Determination and Control

Since there are multiple deliverables for this spacecraft, we will be using multiple sensors to satisfy different requirements. When making the decision on what sensors to choose, mass, power, precision, and cost were taken into account. These parameters were the ones considered because the sensors need to be extremely precise but also not overly heavy and expensive. By looking at the decision matrix for the attitude determination, Table 12, the conclusion of using a star tracker and sun sensors can be made. These two sensors are extremely important to ensure the spacecraft's orientation remains correct and aligned with both the Sun and Earth so the power and telecommunications subsystems can work as intended.

For the decision for which actuator to use, instead of taking cost into account like with the sensor, the speed/reaction is valued more. The speed at which the actuator can respond is important because this is what controls the error of our system. Error is created from the time difference in gathering the data to reacting and physically changing orientation, so a fast reacting actuator will lower the error, which is ideal for the spacecraft. As seen in Table 13, the use of reaction wheels is best for our desired system. [31] [34]

	Weight	Star Tracker	IMU	Sun Sensor	Magnetometer
Mass	2	2	1	3	3
Power	1	3	5	3	5
Precision	2	5	4	4	2
Cost	3	2	2	3	2
Total	40	23	21	26	20

Table 12: Decision Matrix for Attitude Determination Sensor

	Weight	Reaction Wheel	Reaction Thruster	Gyros	Magnetorques
Mass	2	3	3	3	4
Power	1	2	1	2	2
Speed	2	3	2	2	2
Precision	3	3	2	2	2
Total	40	21	17	18	20

Table 13: Decision Matrix for Attitude Control Actuator

7.3. Non-technical Considerations

7.3.1. Public Health and Safety

In testing, there is potential safety concern due to the fast rotating control components which can cause harm. Launch failures can pose risk to public health

7.3.2. Global

The Attitude Determination and Control subsystem does not have any immediate global concerns.

7.3.3. Cultural

The Attitude Determination and Control subsystem does not have any immediate cultural concerns.

7.3.4. Social

The Attitude Determination and Control subsystem does not have any immediate social concerns.

7.3.5. Environmental

This mission could unintentionally add to space debris as well as harm the Europa environment by contamination

7.3.6. Economic

Attitude Determination and Control although is not a major financial load, making sure there is no waste with resources. Delay in the mission can also cause financial cuts.

7.3.7. Professional Ethics

There are no immediate concerns for professional ethics for Attitude Determination and Control because every component is selected with the mission's interests.

7.4. Risk Assessment

The mission heavily relies on an exact measurement and analysis of attitude, so the orientation of the spacecraft and the components are facing the correct direction and at the most ideal angle. Since the spacecraft is very reliant on the success of the ADCS subsystem, the risk severity is very low and the risk mitigation for the system is important.

Reaction wheels are placed in three different directions in order to have three-axis control. If one of the wheels fails though the entire system fails since there cannot be torque added in the respective axis of the failed wheel. To mitigate this risk, another wheel can be added to the system either along tetrahedral axes or as a spare to the three-axis control. In both of these cases if one wheel fails the extra fourth wheel will still be able to apply torque in the axis. [31] [34]

Another main risk of the system is in the sun sensor, if the sun sensor fails the spacecraft will not be able to track the sun and maintain its orientation. Just like the risk mitigation with the reaction wheels, to mitigate the sun sensor failure, more sensors can be added as a backup. This risk mitigation is shown in tabular form below in Table 14.

Risk	Mitigation	Severity
Reaction Wheel Failure	Adding More Reaction Wheels so if one fails the other reaction wheels can make up for the failed wheel	1
Sun Sensor Failure	Add More Sun Sensors, if the sun sensors fail and can no longer track the sun, adding more sun sensors to cover the areas that aren't by the original ones	1

Table 14: Risk Mitigation for ADCS

7.5. Future Work

For the final report, the torque applied to the system and frequency of the noise needs to be calculated. Error in the system will exist since the spacecraft cannot react immediately, it takes the sensors time to analyze the data and then move based on the analysis. Focus on calculating the attitude now that we have chosen the sensor and actuators for the spacecraft.

8. Mechanisms and Deployables (Nick)

8.1. System Definition

An important element carrying out the EO mission is the Mechanisms/Deployables subsystem which integrates all movable and mechanical parts of the spacecraft. This subsystem performs functions of deployment of the scientific equipment, the restructuring of the components of the spacecraft stage-wise and the general functionalities of the spacecraft throughout its movement and operational tasks around Europa. The effective launch of the mission is dependent on the successful development and combining of this subsystem within the overall system. The objectives of the Europa Orbiter mission are successfully achieved by the Mechanisms/Deployables subsystem which has the responsibility of effective data gathering, accurate positioning of instruments onboard the spacecraft, and maneuvering of the spacecraft as required by the mission.

8.2. Key Components and Functions

The mechanisms and deployables subsystem encompasses the EO's mechanical and moving parts. The main duties of this subsystem include the deployment of scientific instruments, the positioning of autonomic devices, including antennas, solar panels, and instruments, as well as performing various mechanical functions, such as rotation and robotic movements.

8.2.1. Probe Deployment

The design of the probe deployment is essential for the success of the mission. There are three options being considered: a system that is spring loaded, a system that is driven by an electrical motor and the use of latch systems [45]. In spring loaded systems, springs or similar mechanisms are used to ensure that the Minos Probe has the confidence of being deployed. The simplicity and reliability of spring-loaded systems make them the most preferred option. The systems that are driven by electrical motors are good to use when control over the release process is essential, since it greatly reduces impact forces at landing. Such systems are able to perform simple and effective movement of mechanical parts and therefore do possess a lower power load. Latch mechanisms will be used to lock the probe in place during motion

and then released when the desired time comes to activate the next step of the mission. These systems make sure that the probe is stable and it is well protected until such time that deployment is required.

	Weight	Spring Driven	Motor Driven	Hybrid
Required Power	2	5	1	3
Reliability	1	3	4	5
Accuracy	3	2	3	4
Complexity	2	4	2	5
Mass	2	3	1	2
Total	50	33	21	37

Table 15: Decision Matrix for Probe Deployment

8.2.2. Deployment Mechanisms

A key component of the mission's scientific goals is the placement of research instruments. The precise placement of all scientific devices for the best possible data collection is guaranteed by the Mechanisms/Deployables subsystem. This entails the use of precision positioning devices and extendable booms that may change the distance and orientation of instruments like spectrometers and cameras. The subsystem improves the quality and dependability of the scientific data collected by preventing contamination or interference from the spaceship body by putting instruments on extendable arms. For example, Hooke's Law can be used to determine the force needed to extend an instrument on a boom, and the result is as follows:

$$F = kx$$

where x is the displacement from the equilibrium position, F is the force applied by the spring, and k is the spring constant.

8.2.3. Antennas and Panels

Configuring a spacecraft entails modifying a number of parts to accommodate various mission stages. This covers the installation and modification of sensors, solar panels, and antennas. In order to accommodate vehicle limitations, high-gain antennas are compressed during launch. Once in orbit, they are expanded to enable high-rate data transfer. NASA's Lunar Reconnaissance Orbiter (LRO) instruments deployments show how multiple sensors and other deployables are integrated uniquely [43]. The following formula can be used to determine the torque needed to extend a hinged antenna:

$$\tau = r \times F$$

where F is the force applied, τ is the torque, and r is the distance between the pivot point and the force application point. This guarantees that the antennas are positioned precisely and with the required force.

Low-gain antennas provide vital data linkages for the duration of the mission by guaranteeing constant communication with Earth and the EO spacecraft. During launch, solar panels are intended to stay folded. It is possible to estimate the electricity produced by the solar panels using:

$$P = A \times r \times E$$

From this equation, P is the power generated, A is the solar panel area, r is the local sun irradiance, and E is the solar cell efficiency. To maximize solar exposure and reduce radiation, the panel must be optimized with adjustable angles.

8.2.4. Arm Mechanisms

An alternative approach is to integrate a robotic arm for the mission. The design will provide more precise results with the cost of added weight. This mechanism would be better suited for covering the movement of any robotic components on board as well as orientation modifications to maintain the proper attitude for instrument operation and communication. The arm would have hinged a simple claw as its end effector, offering two degrees of freedom to execute complex movements. These mechanisms would allow for a wider range of movements and could be executed with human interaction.

This design stands out as the most dependable due to its heritage and reliability. However, The extended length of the arm adds a large moment of inertia, and similar to the antennae, the resulting torques from arm movements could be a factor in stability. Along with the torque equation from before, Newton's second law can be used to model the dynamics of these movements:

$$F = ma$$

where m is the mass of the moving part, a is the acceleration, and F is the applied force. This guarantees the seamless and accurate execution of all mechanical activities, preserving the spacecraft's functionality and integrity throughout mission operations.

	Weight	Articulated	Linear	Modular
Required Power	2	2	4	2
Reliability	1	4	5	5
Accuracy	3	1	3	2
Complexity	2	4	2	5
Total	40	19	26	25

Table 16: Decision Matrix for Arm Mechanisms

8.3. Design Requirements

8.3.1. Reliability

A key emphasis of the mechanisms and deployables subsystem is reliability as fail-safe mechanisms are incorporated that can support the system to help provide deployment even with system failures. Actuators could be employed for critical deployment tasks to prevent single points of failure, and alleviating too many dependencies which would improve the overall effectiveness of the subsystem.

8.3.2. Mass

Limitations on bulk and volume are a major challenge. To keep the subsystem as light as feasible overall, all materials must be robust but lightweight. Additionally, all deployable systems are made to fit inside the spacecraft's structural boundaries both during launch and travel, saving space without compromising functionality. Simply said, the subsystem's weight is the total of the weights of its constituent parts.

8.3.3. Power

Controlling power usage is essential to the effectiveness of the mission and in order to prioritize deployments and guarantee that functions receive enough power, the subsystem is coupled with the spacecraft's power subsystem. Mechanisms are made to run on very little power, preserving the spacecraft's power budget and enabling longer mission durations. Each deployable mechanism's power consumption can be computed using:

$$P = IV$$

from the equation, P is power, V is voltage, and I is current.

8.3.4. Material

The subsystem's materials must be able to endure the harsh environment of space and fluctuating circumstances. Materials that can endure the intense cold of space and the fluctuating thermal conditions close to Jupiter must be used to make components. To ensure long-term performance, components are either insulated or built from

radiation-hardened materials to remain active with Jupiter's severe radiation belts. The thermal resistance of materials can be assessed using the heat transfer equation:

$$Q = mc\Delta T$$

m is the material's mass, c is its specific heat capacity, ΔT is the temperature change, and Q is the heat absorbed or released. This process helps guarantee that the materials chosen match their ability to function while maintaining structural integrity while withstanding temperature variations.

8.4. System Integration

8.4.1. Data

Effective integration with the spacecraft's command and data handling systems ensures precise control and monitoring of the Mechanisms/Deployables subsystem. Deployment mechanisms are seamlessly integrated with the spacecraft's command systems, allowing for precise control over all deployable components. Additionally, sensors are implemented to monitor the status of deployable components in real-time, providing mission control with continuous feedback and enabling prompt responses to any issues. The data flow can be modeled using information theory principles, ensuring that telemetry data is transmitted accurately and efficiently without loss or corruption.

8.4.2. Power

Integration with the power subsystem ensures that all deployable mechanisms receive adequate power without compromising the functionality of other subsystems. Sufficient power is allocated to operate all deployable mechanisms, and energy efficiency is prioritized to operate within the mission's power budget constraints. Power distribution networks are designed to handle the dynamic demands of deployment tasks, and energy storage solutions are implemented to provide reliable power during critical operations.

8.4.3. Thermal

Being able to Control the temperature will play a crucial role in the success of this subsystem's operations and to guarantee that everything operates within varying temperature ranges precautions must be made and material testing to avoid overheating and any damage, devices are also built to control and disperse heat produced during deployment activities.

8.5. Variables

8.5.1. Environmental

The Mechanisms/Deployables subsystem faces immense obstacles when operating in Jupiter's harsh atmosphere. Deployable systems must be built with strong radiation

resistance to ensure they continue to work in spite of Jupiter's powerful radiation belts due to the high radiation levels in the vicinity of the planet. Furthermore, mechanisms need to be shielded from potential harm from space debris and micrometeoroids, which could jeopardize their dependability and operation.

8.5.2. Operation Precision

Achieving precise deployment and configuration of the spacecraft's components is critical for mission success. Mechanisms must ensure that scientific instruments are accurately positioned for optimal data collection and that antennas and solar panels are deployed correctly to facilitate communication and power generation. Real-time data about the spacecraft's orientation and environmental conditions may necessitate adjustments to deployment actions, requiring mechanisms to be highly responsive and adaptable.

8.6. Ethical Considerations

- 8.6.1. One of the key risks that should be considered is the environmental impact the deployables could have on Europa, whether things go wrong or right. Another key component to keep in mind is The Committee on Space Research (COSPAR) imposes specific protocols for sterilization and cleanliness to prevent biological contamination of planetary bodies. Deployables—especially components like the Minos Probe that will make contact with Europa's surface—must also be sterilized according to these standards to minimize any biological contamination risk. Adhering to (COSPAR's) protocols is also the ethical decision in ensuring that future space exploration missions can be successful. [44]

8.7. Future Steps

Thorough testing and validation processes are necessary to guarantee that the Mechanisms/Deployables subsystem operates as planned. To confirm their robustness and functionality in space-like circumstances, mechanisms are put through vibration, radiation, and thermal vacuum testing. Vibration testing replicates the forces encountered throughout launch, while thermal vacuum tests mimic the intense cold and vacuum of space. In order to ensure all deployable components function properly during real mission operations, many deployment tests are carried out to evaluate their accuracy and dependability.

9. Thermal (Alexis)

9.1. System Definition

Given the journey from Earth to Jupiter, the spacecraft will have to traverse through many conditions in space; including, but not limited to, the heat from the Sun and the

coldness that comes from the vacuum that is space. To prevent damage to the spacecraft and minimize fatigue of components over time, it is important to consider these conditions the spacecraft will go through in space and address them when possible[18].

9.2. Initial Selection

Though passive thermal protection is ideal, as it does not require power consumption, it does not provide enough thermal control; hence both active and passive thermal regulation will be utilized. The selections relating to thermal regulation will be split up into the active ways the spacecraft regulates its temperature and the passive ways it will regulate the temperature of the internal and external components.

9.2.1. Active Thermal Control

To get the internal components to be in an environment where they can function, heat needs to be controlled; the best way to do this is through active thermal regulation. The active thermal regulation will have two parts; the first is a way to generate heat and the other is a way to manage heat. Though radioisotope heating units (RHU) are passive, it will be considered in active thermal regulation methods as it would be coupled with a way to manage the heat produced. As such, the decision between resistive heaters (RH) and radioisotope heating units is necessary, to choose in which way to heat the spacecraft and can be seen in Table 17 [35][36].

	Weight	RHU	RH
Cost	1.0	3	5
Durability	1.5	5	3
Mass	1.5	3	4
Environmental Concerns	0.5	1	3
Heat Generation	1.5	4	4
Flight Heritage	0.5	5	5
Power Consumption	2.0	5	2
Technical Readiness	1.0	5	5
Volume	0.5	2	4
Total	50	40	36.5

Table 17: Decision Matrix for Heat Generation

The radioisotope heating unit seems to be a better fit; this is no surprise, as a spacecraft on a similar mission, the Europa, used a radioisotope heating unit as well[38]. The active thermal regulation methods will be used to manage the amount of heat the internal components will receive. The methods up for consideration to manage heat are a heat redistribution system (HRS) and thermoelectric heat pumps (TEHP); the evaluation of which can be seen in Table 18 [37][38].

	Weight	TEHP	HRS
Cost	1.0	2	4
Durability	1.0	3	4
Mass	2.0	5	4
Environmental Concerns	0.5	3	5
Flight Heritage	0.5	3	5
Power Consumption	2.0	1	5
Technical Readiness	1.0	3	5
Thermal Control	1.5	4	3
Volume	0.5	5	3
Total	50	31.5	42

Table 18: Design Selection Matrix for Active Thermal Control

In the end, the heat redistribution system fits best with the mission.

9.2.2. Internal Passive Thermal Control

Since passive thermal regulation does not require power, it is possible to consider a combination of internal thermoregulation methods; though it is important that they complement each other. Of course, there are other factors to consider; hence why three different internal passive thermal regulation methods, thermal straps, foam insulation, and radiator and louver, are compared in Table 19 [38][39].

	Weight	Thermal Straps	Foam Insulation	Radiator and Louver
Cost	1.5	4	5	3
Durability	1.0	5	3	4
Thermal Management	2.0	4	3	5
Mass	2.0	3	5	3
Environmental Concerns	1.0	4	1	4
Flight Heritage	1.0	5	4	5
Technical Readiness	1.0	5	3	4
Volume	0.5	4	1	3
Total	50	41	35	39

Table 19: Decision Matrix for Internal Passive Thermal Control

Given the environmental concerns, outgassing, and the amount of volume it takes up, foam insulation will not be used. Because of that, the mission will use thermal straps and a radiator and louver to help control thermal on the interior of the spacecraft.

9.2.3. External Passive Thermal Control

As with the passive thermal regulations internally, there can also be a combination of external passive thermal regulations. The difference is, that there is no other thermal regulation externally; so a combination is even more necessary, so that way the external components function properly. The different methods include using a honeycomb structure to distribute the heat from the Sun, a heat switch, surface

coatings, and a multilayer insulation (MLI) blanket[38][37]. Table 20 shows the evaluation of these different methods.

	Weight	Honeycomb	Heat Switch	Surface Coating	MLI
Cost	1.5	3	1	3	3
Durability	1.0	5	2	3	4
Mass	2.0	3	2	5	4
Environmental Concerns	1.0	3	1	3	3
Flight Heritage	1.0	5	3	4	5
Technical Readiness	1.0	5	3	4	5
Thermal Management	2.0	4	5	3	5
Volume	0.5	3	4	5	3
Total	50	38	26.5	37	41

Table 20: Decision Matrix for External Passive Thermal Control

A honeycomb structure, surface coatings, and a multilayer insulation blanket seem to be the best way to control thermal on the exterior of the spacecraft.

9.3. Non-Technical Considerations

Having discussed the technical design factors, there are also a few non-technical considerations that need to be considered. These factors can be seen in Table 21 [37][38][39].

Topic	Consideration
Public Health & Safety	The thermal control methods have many safety concerns, as such the proper care will be taken when handling and for disposal.
Global	The global impact of thermal control is that overtime it is slowly evolving, as researchers from all over work on better methods; and this can help contribute to research. There is also the impact of getting materials globally.
Cultural	Thermal control does not seem to have any significant cultural considerations.
Social	The creation of the honeycomb structure, the multilayer insulation blanket, the radioisotope heating unit, thermal straps, and surface coatings will likely result in work for people.
Environmental	While there are environmental concerns regarding the thermal control components, they were taken into consideration within

	the selection process in order to avoid overly harmful components.
Economic	Some options for thermal control are incredibly expensive, so more cost friendly options were given a slight preference over expensive options. This should make the options chosen relatively reasonable. The buying of these components may have a small impact on the economy.
Professional Ethics	Given professional ethics, thermal control components have been chosen in a way that considers not just technical aspects, but non-technical as well.

Table 21: Non-Technical Risks for Thermal Regulation

9.4. Risk Assessment and Mitigation

Due to thermal playing an integral role to many other subsystems, it is important to consider the risks of the low temperatures, high temperatures, and failure of components having an effect on thermal components, and ways to mitigate these risks. Table 22 can be seen outlining these risks[37][38].

Risk	Mitigation	Severity
Extremely low temperatures	The radioisotope heating unit will be used to heat up the enclosed space and the heat redistribution system will help to control the heat components get, along with the passive thermal straps.	2
Extremely high temperatures	The heat redistribution system will help to control the temperature of the interior, along with the radiator and louver. On the exterior, a honeycomb structure, surface coatings, and a multilayer insulation blanket will be used to minimize the heat from the Sun.	2

Table 22: Technical Risks for Thermal Regulation

9.5. Future Work

The exact specifications of the thermal components will be selected and established in the final design review. To keep the components safe, it will be necessary in the future to calculate the specifics of the thermal margins of the varying subsystems. This will also help to figure out what will be required from the HRS to keep the internal components within their acceptable range to not just survive but function; the heat from the electronics and the passive thermal protections be taken into account when

calculating what will be required of the HRS. The calculations of what conditions each component needs will determine where they will need to be located with respect to the HRS[jones 1]. However, given the time constraint, testing of critical points and more in-depth modeling will not be possible.

10. Power (Alexis)

10.1. System Definition

The purpose of the power system is to design in which ways the spacecraft will generate, store, regulate, and distribute power throughout the spacecraft. This is crucial, as too little power could result in a failed mission and too much power would be inefficient and costly. Each other subsystem relies on having enough power to perform their duties, and can hence say the power is an incredibly important part of the spacecraft.

10.2. Initial Selection

Over the years, space vehicles have varied in the ways their power is generated, stored, and managed to best fit their mission requirements; as such, it is necessary to determine it for this mission as well. These responsibilities will be further discussed to determine the best options for power generation, batteries for power storage, and power management electronics[40]. Each of these decisions will be based on several criteria and are weighted a certain amount depending on their importance.

10.2.1. Power Generation

The first decision to make is based on power generation. Power will be needed over a long period, as the mission will take multiple years. As this mission is planned to require power for over a few years, primary batteries and fuel cells, which only last days to months, will not be considered for this mission. The options to be considered will be solar arrays, radioisotope thermal-electric generators (RTG), and a nuclear reactor (static) [37] [18].

	Weight	Nuclear Reactor (Static)	RTG	Solar Arrays
Cost	1	1	2	4
Environmental Concerns	1.5	2	2	4
Flight Heritage	1	2	5	5
Mass	2	2	3	4
Power Output	2	5	3	4
Safety	1	1	4	5
Technical Readiness	1	2	5	5
Volume	0.5	2	4	3
Total	50	24	33	42.5

Table 23: Decision Matrix for Power Generation

As shown, the best option for power generation for this mission is the solar arrays, and as such they will be used as this mission's way of power generation.

10.2.2. Power Storage

Since solar arrays will be used for power generation, it is important to determine which batteries will be used to store power when the spacecraft is not in direct line with the Sun. There are a range of options, which have been narrowed down to a nickel-hydrogen battery (Ni-H), nickel-cadmium battery (Ni-Cd), and lithium-ion battery (Li-Ion) and compared in Table 24 [41][42][43].

	Weight	Ni-H	Ni-Cd	Li-Ion
Cell Life	2	3	5	4
Cost	0.5	3	4	2
Environmental Concerns	1	4	1	3
Flight Heritage	1	5	5	5
Mass	2	2	4	5
Safety	1	4	1	2
Specific Energy	1	4	2	5
Technical Readiness	0.5	5	5	5
Temperature Range	0.5	4	5	2
Volume	0.5	2	2	4
Total	50	28	25	31.5

Table 24: Decision Matrix for Batteries

Given the criteria, the lithium-ion battery is the best option and will therefore be used as this mission's way of power storage.

10.2.3. Power Management

Even though these were the best options, it is important to do everything possible to keep both the solar array and the batteries within certain conditions to keep them from degrading; hence the batteries will be located within the spacecraft, so they do not

freeze from the cold or rupture from the heat. Thermal protection of some kind may also be needed for the solar arrays, so as not to have as much fatigue. To handle excess power that the batteries can not handle, shunts will be used[18].

10.3. Non-Technical Considerations

Having discussed the technical design factors, there are also a few non-technical considerations that need to be considered. These factors can be seen in Table 25 [37].

Topic	Consideration
Public Health & Safety	The batteries have a safety risk, because of this it is important to handle the batteries with care and to have their surroundings be properly managed.
Global	The global impact of power generation and power storage is that overtime it is slowly evolving, as researchers from all over work on better methods. There is also the impact of getting materials globally.
Cultural	Power generation and power storage do not seem to have any significant cultural considerations.
Social	The creation of solar arrays and batteries will potentially result in work for people.
Environmental	Both power generation and power storage pose environmental concerns. The choices made for the sake of power took these concerns into consideration, which resulted in power generation and power storage that may still impact the environment, but as minimally as possible.
Economic	All options for power generation and power storage were incredibly expensive, though a selection was made that was still expensive, but reasonably so given that it is a spacecraft. The buying of these components may have a small impact on the economy.
Professional Ethics	Given professional ethics, power generation and power storage have been chosen in a way that considers not just technical aspects, but non-technical as well.

Table 25: Non-Technical Considerations of Power

10.4. Risk Assessment and Mitigation

Due to power playing an integral role in many other subsystems, it is important to consider the risks of producing too much power, being unable to generate power, and the ways to mitigate these risks. Table 26 can be seen outlining these risks [37] [41] [42] [43].

Risk	Mitigation	Severity
Too much power produced from solar arrays	Having batteries to collect power and have a source of power dissipation.	2
Battery failure	Have multiple batteries in parallel, within the vault where the conditions are controlled, and have a way to prevent overcharging.	2
Not enough power generated from solar arrays	Reduce causes of outgassing, have strict power scheduling, and have limitations on power.	3

Table 26: Risks, Mitigations, and Severity of Power

10.5. Future Work

The exact specifications of the power generation and storage will be selected and established in the final design review. Once the necessary selections for components have been made, the power consumption of the spacecraft will be calculated and made into a power schedule. Included in this power budget will be the power requirement of components actively in use and the power requirement of being on standby, as well as power priorities and when each component will receive said power. A feature of this schedule should be that it is relatively consistent in the amount of power being consumed; as it is a more efficient use of power, as compared to periods of larger and smaller power usage. When it is determined how much power is needed during the mission, the size of power generation, storage, and the bus voltage, distribution, and regulation aspects can be calculated.

11. Command and Data (Kai)

11.1. System Definition

The Command and Data subsystem is responsible for all actions occurring on the spacecraft. It collects data from the scientific instruments on board and processes data sent to the ship from Earth. This subsystem will be responsible for the execution of

commands built into the computer and commands sent from Earth while on its flight path. It will also be responsible for sending the payload and engineering data back to Earth accurately and consistently [18]. This mission will require an extensive amount of data to be sent to Earth due to the nature of the exploration which includes high-resolution images of Europa, collection of Europa probe data, and spacecraft system data to ensure the stability of the spacecraft.

11.2. Initial Design Requirements

From the mission objective, the team was able to create a list of requirements that the Command and Data subsystems must meet to be successful. The first requirement was that it must have redundancy, meaning that there are multiple computers allowing for fewer single-point failures in this subsystem that would jeopardize the mission. The technology must be able to handle radiation since it will be exposed to the harsh radiation environment of Europa. The next requirement is that the technology can handle the vibrations at launch without breaking. The technology must be able to process data from scientific instruments and send that data to Earth efficiently. The last requirement is that the technology must be able to receive commands from Earth and distribute them to the right subsystems [24]. The mass for a subsystem in the spacecraft should always be aimed to minimize where it can, the power usage for the computer is constant but does have a range based on the computer chosen.

11.3. Computer Selection

From the initial design requirements, the team was able to set a baseline for the minimum required technological specifications to choose a computer architecture that would satisfy the mission objectives. The team had to choose whether to use a centralized processing or a distributed processing unit [25]. The decision matrix selection for those two types of computer architecture can be seen in Table 28.

	Weight	Central	Distributed
Weight	3	4	2
Volume	3	4	2
Power	3	4	2
Reliability	4	2	4
Cost	1	3	2
Risk	2	2	3
Total	80	51	42

Table 27: Decision Matrix for Computer Architecture

The team decided to use a centralized processing unit since it is more suitable for the time frame of this mission and minimizes areas of interest such as mass and power consumption than a distributed processing unit. With all aerospace technology, the risk of space radiation affecting equipment is present and must be mitigated to allow for the accurate data collection of all scientific instruments on the spacecraft.

11.4. Non-Technical Considerations

11.4.1. Public Health and Safety

The Command and Data is responsible for making sure the programmed information is executed such that the launch into space and the flight trajectories are successful. If that does not happen, then the computer also has successfully integrated a system where in such events there is a back up plan to not affect the public such as redirecting to a proper space dump area [26].

11.4.2. Global

The Command and Data subsystem does not have any immediate global concerns.

11.4.3. Cultural

As a culture, people see hope in space exploration for the future of the species. This hope can only be entertained by obtaining the data necessary to prove the success of the mission. The Command and Data is responsible for ensuring the safety of this data.

11.4.4. Social

The Command and Data is responsible for executing the commands sent to it from Earth. It also is responsible for maintaining the trust of the public along with the engineers involved in designing it. To keep good opinions about space exploration, the computer must successfully bring the spacecraft to its desired destination and retrieve the data it was sent for.

11.4.5. Environmental

The Command and Data subsystem does not have any immediate environmental concerns for Earth or Jupiter.

11.4.6. Economic

The Command and Data subsystem along with all the other components of the spacecraft are extremely expensive due to manufacturing costs and the need for high reliability. With the failure of this mission comes a heavy toll on the economic status

of the customer and engineers responsible for the mission. The Command and Data subsystem specifically can be expensive when considering the quantity of data storage needed and the processing speed required to quickly get data transferred.

11.4.7. Professional Ethics

The Command and Data subsystem does not have any immediate professional ethics concerns.

11.5. Risk Assessment and Mitigation

The importance of the Command and Data System for this mission are high as the objective is to be able to get enough data to confirm the hypothesis about Europa. From preliminary analysis of the environment the spacecraft will be traversing, the radiation and magnetic field can be detrimental to the stability of the Command and Data subsystem. The radiation and magnetic field from Europa's atmosphere and Jupiter can affect the data storage and other technology required to complete the mission successfully because an exposure that lasts too long can lead to memory corruption and irreversible damage to the technology onboard. Another risk that is relevant before leaving Earth are the intense vibrations the spacecraft will endure while getting out of Earth's atmosphere. These vibrations can permanently damage the technology and leave it useless. Understanding these risks allowed the team to come up with a solution which was to secure the electronic components that cannot be affected in a vault that is less permeable to radiation and magnetic disturbances. The electronics in this vault have the tiny spaces between them filled with a substance used to increase the damping such that there is less stress on the components. These are the immediate risks and plans to mitigate them, in the case of something happening during the mission, the team will quickly respond to resolve the problem if solvable. Table 28 shows the risk severity comparison with the mitigation required to successfully complete the mission.

Risk	Mitigation	Severity
Radiation	Use radiation hardened electronics so they can withstand the harsh environments in space	4C
Vibrations	Apply substance between spaces in electronic components to increase damping and decrease stresses on electronics during launch	3C

Table 28: Technical Risks for Command and Data

11.6. Future Work

For the Final Design Report, the Command and Data subsystem will have a specific design for the computer mass data storage assortment. This will include decisions on what type and how many of the storage devices will be present on the spacecraft and how the processing of data will be completed.

12. Telecommunications (Kai)

12.1. System Definition

The Telecommunications subsystem is critical to ensuring that communication between Earth and the spacecraft is efficient. This subsystem is responsible for determining the size, number, type, and nature of the antennae on the spacecraft. It is also responsible for choosing the frequency bands with which the Earth and the spacecraft will communicate when uplinking and downlinking data [18].

12.2. Initial Design Requirements

For this mission, it is imperative that Earth is able to efficiently send and retrieve data from the spacecraft. Since the mission entails traveling far from Earth, the antennae must be able to send transmissions a minimum of the maximum distance the spacecraft will be from Earth at any given time. These transmissions will vary on importance level dependent on what data is being transmitted which will determine what frequency band to communicate on at the given times. The spacecraft must be able to communicate with Earth at most times even if it is mainly autonomous in the case of needing recalibration data from Earth to successfully get to its destination and complete commands once it is there. Another important design requirement is that there is redundancy in this system or few single point failures such that the mission will not have failed if one system doesn't work. If the Telecommunications fail once the spacecraft is at its destination point, then there is nothing the team can do to retrieve the data it was sent for. This means that multiple antennae must be present on multiple connection points to ensure the reliability of the subsystem. When choosing the antenna, it is important to optimize the parameters such that power consumption is low, mass is low, transmission rates are high and general size is low. Another important requirement for antennae are the thermal considerations. Since the antennae need to be on the outside of the spacecraft, they need to be able to handle temperature changes and variations in radiation exposure. [27]

12.3. Antennae Selection

When choosing the antennae for a mission, it is usually an array of antennae that are selected since they are needed for different features during the mission. It is customary to attach at least one High Gain Antenna (HGA) and an assortment of Medium Gain Antennae (MGA) and Low Gain Antennae (LGA) to the spacecraft. The most important one is the HGA as this is the primary antenna used to communicate with Earth. The selection process consists of choosing the data interface, modulator, power amplifier, and antennae geometry type. In this report we will look at the geometry of the main HGA and the other antennae in the future. In Table 29, the different types of geometry are placed in a decision matrix to decide which configuration to pursue. This will also help decide what antennae configuration would be useful on the other antenna on the spacecraft. [28]

	Weight	Turnstile	Horn	Helix	Bi-Cone	Parabola
Weight	3	5	3	4	2	2
Size	3	5	4	3	1	2
Power	2	5	3	3	4	3
Reliability	4	1	1	1	1	1
Cost	1	5	4	2	1	3
Peak Gain	5	1	3	4	2	5
Beam Width	5	1	3	4	2	5
Risk	2	1	1	1	1	1
Total	125	61	67	75	44	77

Table 29: Decision Matrix for Antenna Configuration

The risk and reliability of these configurations are the same because these configurations have the same technology readiness level since they have been used and tested before. The radiation will affect the antennae the same and all antennae will need proper protection from such radiation. The important aspects are “Peak Gain” and “Beam Width” because those are the factors that control how far the spacecraft can travel and stay in communication with Earth. The rest of the decision factors are important as well but with an increased mass budget and power supply the team does not need to worry about them as much. The grading for this matrix does not show the absolute efficiency of these configurations as they are applied where their functions serve a purpose. For the HGA the parabolic configuration is the best choice for deep space communication.

12.4. Non-Technical Considerations

12.4.1. Public Health and Safety

The Telecommunications subsystem does not have any immediate public health and safety concerns.

12.4.2. Global

The Telecommunication subsystem does not have any immediate global concerns.

12.4.3. Cultural

The Telecommunication subsystem does not have any immediate cultural concerns.

12.4.4. Social

The Telecommunication subsystem does not have any immediate social concerns

12.4.5. Environmental

The Telecommunication subsystem does not have any immediate environmental concerns.

12.4.6. Economic

The Telecommunication subsystem has a lot of money poured into its success due to the mission objective revolving around retrieving and sending data. If the components fail there will be a lot of money wasted and needs left unmet.

12.4.7. Professional Ethics

The Telecommunication subsystem does not have any immediate professional ethics concerns.

12.5. Risk Assessment and Mitigation

The Telecommunications subsystem is extremely important to make this mission successful. The objective of the mission is to collect data from Europa and Jupiter and send it back to Earth. If the Telecommunication subsystem does not work properly, then the mission has failed. The major risks for the telecommunication system are the radiation exposure and thermal changes. These risks will be mitigated by using radiation hardened materials such that they should not malfunction while exposed to the radiation of Europa and other space bodies within a certain time range. To protect against thermal fluctuations in space, there are thermal control coatings (TCC) that are applied to the sensitive electronics to help regulate their temperature. In Table 30, the risk is broken down by severity and mitigation required to successfully complete the mission.

Risk	Mitigation	Severity
Radiation	Use radiation hardened electronics such that the exposure from space environments won't harm it with infrequent exposure	4C
Thermal Changes	Use TCC on sensitive electronics to minimize temperature fluctuations	4C

Table 30: Technical Risks for Telecommunication

12.6. Future Work

For the Telecommunications subsystem, specifics for the size and number of antennae on the spacecraft still needs to be decided. The frequency band that each antennae will communicate on also needs to be decided. For these decisions, there will need to be an analysis of the directional antenna gain and the creation of an antenna radiation pattern diagram.

13. Conclusion (Ava)

The Hellotis mission aims to successfully establish a highly eccentric orbit around Jupiter and fly by Europa several times to gain knowledge on its surface from onboard instrumentation and from deploying the Minos probe onto the moon's surface. Future work with Hellotis will involve choosing specific instruments that will be used to analyze Europa, as well as a more in-depth analysis for every subsystem. These analyses will finalize design choices and give quantitative performance estimates to achieve mission success. A final product with detailed analyses will be delivered to the customer in the final design report in December.

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14. Appendices (Ava)

14.1. Summary of Non-Technical Considerations

The chart below provides an overview of the team's non-technical considerations. It highlights that cultural factors were deemed negligible, primarily because the mission takes place in space, where public interaction is minimal. As a result, the impact on culture is expected to be minimal throughout the mission.

Section	Structures	Orbital Mechanics	Propulsion	ADCS	Thermal	Power	Telecom
Public Health & Safety	y	y	y	y	y	y	n
Global	n	n	n	n	y	y	n
Cultural	n	n	n	n	n	n	n
Social	n	n	n	n	y	y	n
Environmental	y	y	y	y	y	y	n
Economic	y	n	y	y	y	y	y
Professional Ethics	y	y	y	y	y	y	y

14.2. Team Structure

Mission Subsystem	Primary Lead	Secondary Lead
Structures	Keval	Ashe
Mechanisms and Deployables	Nick	—
Propulsion	Robert	Alexis
Flight Mechanics	Ashe	Ava
Attitude Determination and Control	Ava	Robert
Thermal Management	Alexis	Ava
Power	Alexis	Kai
Command and Data	Kai	Ashe
Telecommunication	Kai	Nick

14.3. Gantt Chart

[illegible]

14.4. Statement of Work

Mane 4250-01 Space Vehicle Design

Statement of Work

Hellotis: Searching for a New Home

**Team 2: Ava Goucher, Alexis Jones, Ashe Mlodzienski, Robert Quevedo, Keval Shah,
Kai Silva, Nicholas Walker**

Introduction

Europa is one of Jupiter's 95 moons, which scientists from Tuscan, Arizona's Kitt Peak National Observatory have speculated to contain liquid water since 1972 through spectroscopic observations. This discovery prompted more investigation into Europa. NASA sent Voyager 1 and 2 past Jupiter to take clearer pictures of Europa in 1979, which further confirmed the hypothesis of the scientists investigating liquid water on its surface. This led scientists to have high expectations for the Galileo mission that entered Jupiter's orbit in 1995 and had 12 close flybys with Europa while sending data of its surface back to be analyzed (Alexander 2009). The Hubble Space Telescope later detected geyser plumes on Europa's surface in 2012. These findings have led scientists to push for more in-depth investigations because of the potential for the habitability of Europa (Roth 2013).

This project will design a spacecraft to travel to Europa and bring the Minos Probe to study its surface by successfully landing on Europa's surface. This mission will reveal information about the water's formation and ability to stay liquid as well as if there are historic or current signs of life (Pappalardo 2024).

Objectives

The primary objective of the mission is to send a spacecraft to Jupiter, where it will maintain an orbit that allows for close flyby orbits of Europa. This spacecraft will bring with it a suite of nine scientific instruments to study the surface and atmosphere of Europa (NASA 2024) as well as the Minos probe that will land on Europa to collect more data. The spacecraft will also act as the first relay point for Minos' information on its relay back to Earth.

Some potential secondary objectives of the mission, once the primary mission is completed include

- Detailed surveys of Jupiter's other moons
- Gathering information about local variations in Jupiter's gravitational field and possible sources of the North-South asymmetry

Scope of Work

The team will design a spacecraft capable of carrying out a mission to Europa. The work will include the design, analysis, and specification of the following major subsystems:

1. Structures
2. Mechanisms/Deployables
3. Propulsion
4. Flight Mechanics
5. Attitude Determination and Control
6. Thermal Management
7. Power
8. Command and Data
9. Telecommunications

In addition, the design will include space to hold the probe which will be dropped off on Europa on its first flyby.

Deliverables

The following items will be presented to the customer:

- Preliminary Design Review (PDR) Report - 18 October 2024
 - Detailed mission architecture and objectives
 - Decision matrices for each subsystem to detail tradeoffs and design decisions
 - Preliminary analyses of subsystem design and integration and orbital trajectory
 - Risk analysis and mitigation
 - Plan of precession
- PDR Presentation - 22 October 2024
 - Poster presentation on the findings of the PDR Report
- Final Design Review (FDR) Report - 6 December 2024
 - Detailed mission architecture and objectives
 - Detailed designs based on the decision matrices found in the PDR
 - More in-depth analyses of the subsystem design and integration
 - Risk analysis and mitigation
 - Future work needed before launch
- FDR Presentation - 10 December 2024
 - Poster presentation on the findings of the FDR Report

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